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Research Article

Design and Implement Deepfake Video Detection Using VGG-16 and Long Short-Term Memory

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This study aims to design and implement deepfake video detection using VGG-16 in combination with long short-term memory (LSTM). In contrast to other studies, this study compares VGG-16, VGG-19, and the newest model, ResNet-101, including LSTM. All the models were tested using Celeb-DF video dataset. The result showed that the VGG-16 model with 15 epochs and 32 batch sizes had the highest performance. The results showed that the VGG-16 model with 15 epochs and 32 batch sizes exhibited the highest performance, with 96.25% accuracy, 93.04% recall, 99.20% specificity, and 99.07% precision. In conclusion, this model can be implemented practically.

1. Introduction

The artificial intelligence (AI) and machine learning (ML) technologies have grown exponentially. In recent years, they have become more accessible and easier to use, indicating their continued expansion and growth. AI is widely applied in various fields, including medicine [1], education [2], business [3], industry [4, 5], and social media [6]. However, the risk of criminal exploitation increases as the capabilities and applications of AI technology expands. Caldwell et al. [7] conducted a study to identify the potential criminal applications of AI and related technologies that resulted in the construction of a taxonomy comprising approximately 18 threat categories. One of the most concerning emerging hazards is the spread of deepfake technologies.

Deepfakes [8, 9] use AI and ML techniques to seamlessly merge and overlay existing footage onto original images or videos, thereby creating nearly indistinguishable, deceptively realistic content. Recent technological advancements have simplified the creation of deepfake videos that replicate human faces [10] and mimic speech, movement patterns [11], and tones of voice with remarkable accuracy [12]. Consequently, these manipulated videos have entered social media platforms that play an ever-growing role in daily communication and information sharing. Although certain deepfake content is created for entertainment purposes [13], the potential for malicious use is considerable. Misuses of deepfake include impersonation scams such as criminals posing as law enforcement officers during video calls to deceive victims into transferring money [14]. In addition, a Europol report [15] highlighted the potential illicit applications of deepfake technology, including CEO fraud, evidence tampering, and creation of nonconsensual pornography, highlighting the urgent requirement for effective detection and prevention measures.

Techniques such as generative adversarial networks (GANs), convolutional neural networks (CNNs), and common fake feature networks (CFFNs) are widely used to detect deepfakes. The complexity of video data compounds the difficulty of detecting deepfakes in real time and with high accuracy in video streaming. This study examined the complex problem of detecting deepfakes in video formats that are frequently encountered on social media platforms. The transfer learning model and long short-term memory (LSTM) networks are well suited for addressing the temporal aspects of video data. The LSTM methods in combination with VGG-16, VGG-19, and ResNet-101 on Celeb-DF video dataset were determined. This study ensured the continued effectiveness of detection techniques by fostering innovation and diverse manipulation models in the ongoing battle against emerging deepfake threats. The following sections present a detailed account of the methodology used in the study, showing the manner in which the combination of LSTM and established transfer learning techniques can significantly enhance the accuracy and efficiency of deepfake detection. This study emphasizes its relevance in the ever-changing landscape of detecting synthesized video content, underscoring its commitment to be at the forefront of the curve in countering deepfake proliferation.

The remainder of the paper is organized as follows: Section 2 presents a literature review. Section 3 presents the research methodology, including the operational process, dataset, data analysis, and model. Section 4 presents the results of the proposed method and a discussion. The last section presents conclusions and suggestions for future research.

2. Background and Related Works

2.1. Deepfake Detection. Deepfake [16] is derived from two terms: deep learning and fake learning. Deep learning is a branch of AI that learns deeply about the features of a specific person such as face, skin tone, body shape, gestures, movement, speech, and tone of voice [17]. Deepfake refers to the falsification of the identity of a person by AI such as editing the face of a celebrity into a pornographic film, video spoofing in which people speak differently from that in the actual video, or realistic sound spoofing. People who are unfamiliar cannot discern these technologies with the naked eye. The procedure for creating a deepfake begins with the encoder learning to encode the properties of each image such as facial expressions, facial shape, skin tone, and lighting. The encoder output is then presented to the decoder network, where the decoder learns to reconstruct the image by extracting the essential features. After training, the face is swapped, and a new image is created using the properties of the original image. This technique is repeated for each frame to replicate a video.

Deepfake detection involves overlaying and merging images or videos from one batch onto another. Nguyen et al. [18] divided deepfake detection methods into two categories and four subcategories. The first category is fake image detection that includes handcrafted and deep features-based technique subcategories such as generative adversarial networks (GANs), convolutional neural networks (CNN), and common fake feature networks (CFFNs). The second category is fake video detection that is divided into two subcategories: temporal features across video frames and visual artifacts within video frames. Based on temporal features across video frames, this group includes CNN, long short-term memory (LSTM), and long-term recurrent convolutional networks (LRCNs); most methods use the spatiotemporal features of video streams. The video is manipulated frame-by-frame to detect deepfakes. The visual artifacts within video frames subgroup focuses on the approach that decomposes videos into frames and explores visual artifacts within single frames to obtain discriminant features that are then distributed into a deep or shallow classifier to differentiate between fake and authentic videos.

2.2. Transfer Learning. Transfer learning [19] is a machine learning technique that uses a trained CNN as a structure and provides new information for teaching. Currently, transfer learning is typically applied to image classification and prediction such as classifying texts and flower types. In addition, the detection of blinking eyes, smiling, and facial distortion can use different models such as VGG [20], ResNet [21], and Inception-V3 [22]. Each model exhibits different performance depending on the analyzed dataset. In addition, only a few researchers have utilized transfer learning in their experiments. However, no research has compared transfer learning in new formats such as VGG-16, VGG-19 [23], and ResNet-101.

The VGG-16 neural network was developed by researchers in the visual geometry group at Oxford [24]. VGG-16 was among the best-performing architectures in the ILSVRC challenge 2014, with a top-5 classification error of 7.32%. VGG-16 is a transfer learning method of worldwide interest. In most use cases, VGG-16 is used for image recognition or classification [25], image detection and localization [26, 27], and image embedding vectors [28]. The input of the VGG-16 neural network is an image with dimensions of (224, 224, 3). The pooling layer follows the first through fifth convolutional layers with the maximum value (max pooling). The endpoint is then sent to the entire link layer with 4,096 nodes, and the last layer is assigned 1,000 nodes. Results are calculated using the softmax stimulation function.

The VGG-19 neural network [29] has a structure similar to that of the VGG-16 neural network; the VGG-19 neural network consists of 19 layers, irrespective of the top layer. The pooling and 19-layer networks are divided into five groups of convolutional layers. The max pooling layer follows the first through fifth convolutional layers. Then, it is end-pointed to the entire link layer of 4,096 nodes. Subsequently, the final pool of the complete pooling layer is set to 1000 nodes by calculating the result value using the softmax excitation function. The main difference between VGG-16 and VGG-19 lies in the number of layers and variety of images, where VGG-19 has more layers. It is trained using more than 1,000,000 images. It can also group images from more than 1,000 groups, such as people, animals, and objects.

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The residual network is an artificial neural network launched in 2015 by He et al. [30]. A ResNet was created to solve the vanishing gradient problem. This method adds a shortcut to the neural network to bypass layers. Certain parameters are reduced to zero when the model learns by repeatedly updating them during the process. In particular, those in the first layer are closer to the input layer, resulting in a loss function value for the next parameter update. It changed only slightly; therefore, the model was not as ef2ficient as expected. The most widely accepted models in the ResNet family are ResNet-34, ResNet-50, and Res-Net-101. ResNet-101 is a CNN that is 101 layers deep. It can classify images into 1000 object categories such as dogs, cats, and pencils. The network has an image input size of 224-by-224.

2.3. Long Short-Term Memory (LSTM). LSTM [31, 32] is a particular recurrent neural network (RNN) that is capable of learning long-term dependency. LSTM is well suited for classifying, processing, and predicting based on time series data. Moreover, it is more efficient because it contains a standard unit comprising a cell, an input gate, output gate, and a forget gate. A memory cell state can declare which information is essential to remember or eliminate. Within an LSTM network, various functions are used to create a particular function such as data reading, data entry, data updating, and information loss. These functions work similarly to gates for assigning incoming data in the desired direction. LSTM can delete or add data to a cell state that is carefully controlled by this structure. In a cell of an LSTM network, the gate is named forget gate and is used to decide whether the cell should forget the information from the previous timestamp. Following is the equation for the forget gate:

$$f_t = \sigma\left(X_t \times U_t + H_{t-1} \times W_t\right), \tag{1}$$

$$f_t = \sigma\left(X_t \times U_t + H_{t-1} \times W_t\right), \tag{2}$$

where $C_{t-1} \times f_t = 0 \ldots$ if $f_t = 0$ is forget everything and $C_{t-1} \times f_t = C_{t-1} \ldots$ if $f_t = 1$ is forget nothing. X_t is the input to the current timestamp. U_t is the weight associated with the input. H_{t-1} is the hidden state of the previous timestamp. W_t is the weight matrix associated with hidden state. If $f_t = 0$, then the network will forget everything. If $f_t = 1$, then the network will forget nothing.

The other gates are input and output gates. The input gate is used to quantify the importance of the new information carried by the input. The output gate predicts the current output from the hidden state in the hidden state of long-term memory. The formula for these gates is as follows:

$$i_t = \sigma\left(X_t \times U_i + H_{t-1} \times W_i\right). \tag{3}$$

$$o_t = \sigma\left(X_t \times U_o + H_{t-1} \times W_o\right). \tag{4}$$

2.4. Literature Review. Deepfake detection can be accomplished using deep learning methods such as RNN, CNN, and LSTM. In addition, many studies have combined techniques to enhance the accuracy of the model. For example, Stanciu and Ionescu [33] applied facial features and a CNN-LSTM model to implement deepfake video detection. Güera and Delp [34] used an RNN to classify videos subjected to manipulation using various transfer learning techniques [23, 35] for better detection of fake faces. Table 1 lists the research on deepfakes. It can be classified into CNNs and RNNs. CNN can be divided into several types such as VGG and ResNet. Performance testing uses standard metrics such as accuracy rate, precision, recall, *F*1-score, and AUC.

According to Table 1, related research has shown that transfer learning techniques can be used in several ways. For example, Li and Lyu [27] applied VGG-16, ResNet-50, ResNet-101, and ResNet-152 to detect face warping artifacts. Chang et al. [36] used an LSTM and NA-VGG to detect fake faces in videos. In 2021, Rosli et al. [42] presented a comparison 50 multitransfer learning model for a deepfake image recognizer using VGG-19 and ResNet-50. Notably, the use of deep learning techniques in combination with different transfer learning techniques has a high accuracy of 80–100% that varies depending on the dataset used. The commonly used datasets include Celeb-DF [48, 49], DFDC [8], and FaceForensics++ [49]. The current research often combines CNN and RNN techniques. For example, Kaur et al. [39] compared several methods such as face-warping features + CNN, CNN + GRU, and CNN + RNN. Güera and Delp [34] applied a CNN and RNN to the HOHA dataset with high accuracy of 97.1%. According to Elpeltagy et al. [46], the study proposed the CQT XceptionNet and GRU for detecting deepfakes on the FakeAVCeleb multimodal video dataset with 97.52% accuracy.

Spatiotemporal convolutional networks that use spatial information from interframe temporal information also have different uses. For example, Zhang et al. [43] used fixed-length frame volumes collected from a video to feed into a three-dimensional CNN (3DCNN) to detect whether faces were real or fake. A 3D-attentional inception network was suggested by Lu et al. [44] as a method for deepfake detection. De Lima et al. [45] created a performance benchmark for spatiotemporal convolutional methods using the Celeb-DF dataset that included RCN, R3D, MC3, R2Plus1D, and I3D. This study demonstrated that R3D outperformed the other networks. Moreover, Ismail et al. [47] proposed the combination of CNN, HOG, XceptionNet, and GRUs with Celeb-DF and FaceForensics++ (c23). The result of AUROC was 95.53%.

3. Methods

Figure 1 presents an overview of the operation, beginning with providing the video data consisting of real and fake videos to the data preparation phase. In this phase, the video is converted into sequenced image frames. Then, each frame is cropped to show only the face. Subsequently, the prepared data are entered into the model by dividing them into two sets: train and test. The result is a prediction of whether the video is real or fake.

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Table 1: Methodology and performance of related works on deepfake detection.

Method	Dataset	Performance	Reference
VGG network based on noise and image augmentation (NA-VGG)	Celeb-DF	AUC = 85.7%	Chang et al. [36]
XceptionNet and VGG-16	DFFD, UADFV, and Celeb-DF	AUC = 99.67%	Dang et al. [26]
Deep dual-level network	Celeb-DF (v2) and FaceForensics++	Accuracy = 96.5% AUC = 98.9%	Pu et al. [37]
CNN such as VGG-16, ResNet-50, ResNet-101, and ResNet-152	UADFV and deepfakeTIMIT	AUC = 97.4%	Li and Lyu [27]
CNN and RNN	HOHA	Accuracy = 97.1%	Güera and Delp [34]
LSTM and Inception-ResNet-V2	FaceForenscis and Celeb-DF	Accuracy = 91.48%	Yadav et al. [38]
Face warping features + CNN and CNN + RNN	FaceForensics++and CelebDF	AUC = 97.06%	Stanciu and Ionescu [33]
Face warping features + CNN, CNN + RNN and CNN + LSTM	Celeb-DF DFDC ground truth	Accuracy = 96.46%	Kaur et al. [39]
CNN + LSTM	DF-TIMIT UADFV Celeb-DF DFDC preview FaceForensics++	AUC = 99.00%	Rowan and Pears [40]
YOLO + CNN + XGBoost	Celeb-DF-FaceForencics++(c23)	AUC = 90.62% Accuracy = 90.73%	Ismail et al. [41]
VGG-19 and ResNet-50	The first image forensic challenge in 2013	VGG-19: accuracy = 94.65%; ResNet-50: accuracy = 95.08%	Rosli et al. [42]
TD-3DCNN	Celeb-DF	AUC = 88.83% Accuracy = 81.08%	Zhang et al. [43]
I3D att unifAug	Celeb-DF	AUC = 82.78%	Lu et al. [44]
RCN I3D MC3 R3D	Celeb-DF	RCN AUC = 74.87% Accuracy = 76.25% R2Plus1D AUC = 99.43% Accuracy = 98.07% I3D AUC = 97.59% Accuracy = 92.28% MC3 AUC = 99.30% Accuracy = 97.49% R3D AUC99.73% Accuracy = 98.26%	De Lima et al. [45]

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Table 1: Continued.

Method	Dataset	Performance	Reference
CQT XceptionNet and GRU	FakeAVCeleb multimodal videos dataset	Accuracy = 97.52% AUROC = 98.43% Precision = 97.66% Recall = 98.04% *F*1-score = 97.66% Specification = 97.30%	Elpeltagy et al. [46]
CNN + HOG + XceptionNet + GRUs	CelebDF-FaceForencics++ (c23)	AUROC = 95.53% Running time (h) = 13.31	Ismail et al. [47]

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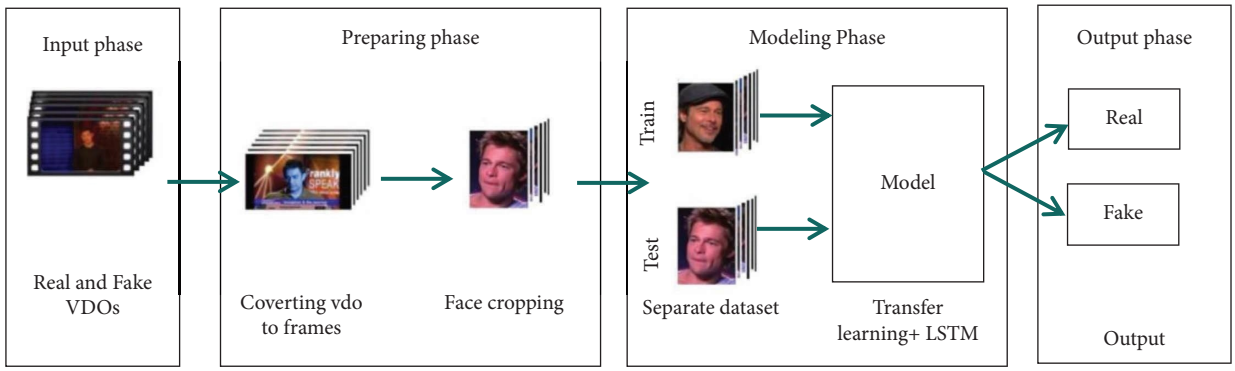


Figure 1: Operation overview.

3.1. Dataset and Data Analysis. This study compiled a collection of Celeb-DF video datasets available at https://paperswithcode.com/dataset/celeb-df. The Celeb-DF video dataset is a large dataset containing both original and deepfake videos. This study generated 240 datasets that were divided into two types: (1) 119 original videos and (2) 121 synthesized videos. The video-to-image sequence and face detection process provided a visual dataset for learning 2,400 images, including actual and fake images. These image collections were then divided into 1,920 learning and 480 test datasets.

To determine the efficacy of the model, we used a confusion matrix, a fundamental and indispensable instrument for machine learning and classification tasks. The primary purpose of this investigation was to determine crucial performance metrics, such as accuracy, recall (completeness), specificity, and precision. The matrix is shown in Figure 2.

Thus,

True positive (TP): The model predicts that the image is a real (1), and the actual label is a real (1).

True negative (TN): The model predicts that the image is a fake (0), and the actual label is a fake (0).

False positive (FP): The model predicts that the image is a real (1), and the actual label is a fake (0).

False negative (FN): The model predicts that the image is a fake (0), and the actual label is a real (1).

3.2. Implementation Environment. In this study, Google Colab was used to implement the aforementioned methods. Using a Turing-based Tesla T4 GPU as the backend of the Google Compute Engine with Python 3.10, two folders were created to hold real and fake datasets. The RAM used was 12.7 GB, and the total disk capacity was 78.2 GB. In addition, a preprocessing file was created to convert VDO and crop faces. Three files were created to implement the model: LSTM with VGG-16, LSTM with VGG-19, and LSTM with ResNet-101.

3.3. VDO Converting and Face Cropping. The video was converted into a sequence of images to detect only the facial area. This study set a frame rate per an image at 0.5 s and limit at ten image frames per video. Figure 3 shows a preview video converted into ten frames.

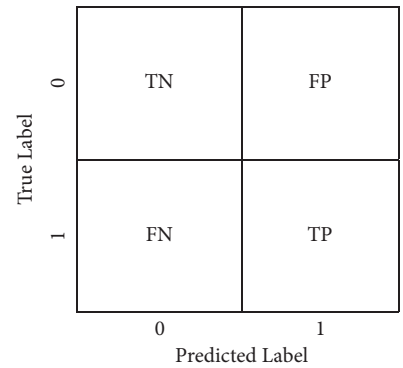


Figure 2: Confusion matrix configuration value.

The next step is face cropping. Captured images were obtained from video sequencing to detect faces and crop the image to obtain only the face area to prepare the image for the next step. The face was cropped using the Python MTCNN library. Figure 4 shows an example of image cropping using the MTCNN library in Python. After video conversion and face cropping, 2,400 images were entered into the model consisting of 1,920 training and 480 test sets.

3.4. Implementing Models. In this study, the VGG-16, VGG-19, and ResNet-101 models were applied in combination with LSTM to analyze deepfake VDO. The objective is to examine the performance of each model and propose modifications for future deepfake videos. Owing to the limited size of each model, all photographs entered were resized to (128×128) dimensions. The entered data were sent to the feature extraction network, where the features were extracted. The extracted features were then categorized using a classification network. Each of these transfer learning models was used to generate the input feature vectors for the LSTM model. Consequently, the classification network was eliminated, and the output vector from the final max-pooling layer was selected as the input vector for the LSTM. The structure of the LSTM model used in this study is shown in Figure 5.

In Figure 5, X_i represents the i-th time step of all the time series. The model used time steps set to a value of 10 and contained a binary classification task with two classes: Class

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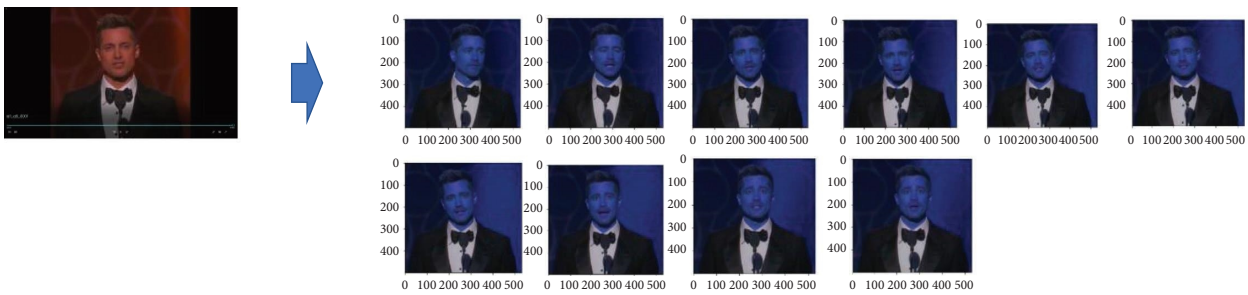


Figure 3: Example of converting VDO to frames.

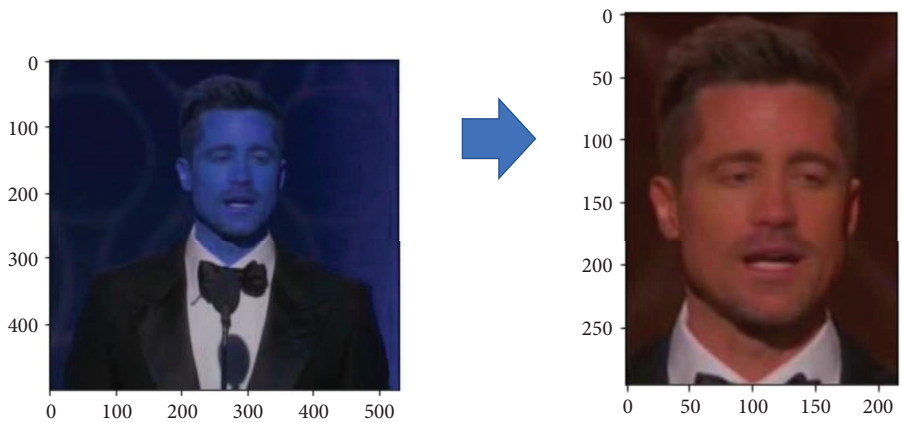


Figure 4: Example of image cropping using the MTCNN library in Python.

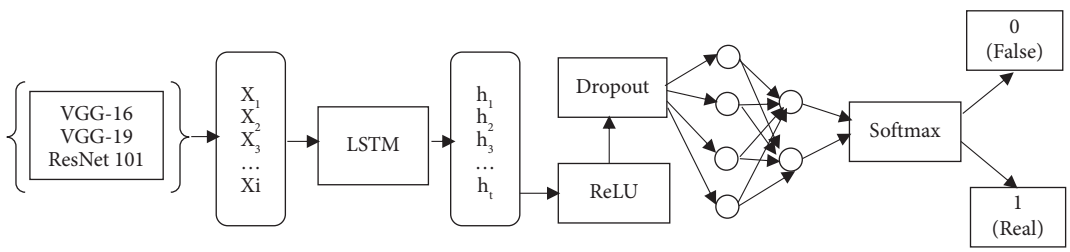


Figure 5: Combination flow between transfer learning models and LSTM.

1 representing the real class and Class 0 representing the false class. The number of hidden units per state was selected such that the model could effectively capture complex shapes while avoiding overfitting. The selected number of hidden units was 1024. The final hidden state, h_t, was then fed into the ReLU function to include a nonlinear relationship between the characteristics of the hidden state. Prior to the fully connected layer, a dropout layer was used with a dropout rate of 0.1 for regularization purposes. Weight decay must not be used in this layer because it may hinder the ability of the model to identify time-dependent patterns. The fully connected layer consisted of two output neurons. These neurons were then subjected to a softmax function that assigns a probability to each class in the output layer. Table 2 presents the parameter configurations used to implement the model.

4. Experiment Results and Discussion

The model performance evaluation was divided into two phases: training and testing. The training phase benchmarked the model using LSTM in combination with transfer learning methods including VGG-16, VGG-19, and ResNet-101 to select the most effective model for detecting fake faces in the next testing phase. This phase defined three evaluation factors: accuracy, loss function, and learning time. The testing phase validated the selected model against the test dataset and used the accuracy, precision, recall, and specificity as the evaluation parameters. During the training phase, the model performance was validated by determining its accuracy, loss function, and learning time to compare the performance of each model, as listed in Table 3.

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Table 2: Initial parameters for implementing the model.

Parameter	Value
Time steps	10
Number of class (real = 1, fake = 0)	2
Input dimension	128,128,3
Dropout rate	0.1
Recurrent dropout	0.1
Hidden layer	Dense(1024, activation = ReLU)
Epoch	15
Batch size	32
Output layer	Dense (number of class, activation = softmax)

Table 3: Results of model performance.

Transfer learning	Epoch	Batch size	Accuracy (%)	Loss function (%)	Learning time (min)
VGG-16 + LSTM	15	32	97.01	8.60	4.22
	15	64	95.38	11.65	8.20
VGG-19 + LSTM	15	32	96.29	9.88	9.45
	15	64	93.23	17.09	4.37
ResNet-101 + LSTM	15	32	42.46	81.32	4.35
	15	64	76.24	50.17	4.13
	50	32	89.32	30.97	13.53
	50	64	88.54	30.86	25.13
	100	32	92.45	19.54	27.08
	100	64	91.02	23.15	25.42

According to Table 3, The VGG-16 combined LSTM technique that was trained with 15 epochs and 32 batch sizes had the highest accuracy of 97.01%, a loss function of 8.60%, and learning time of 4.22 min. The accuracy results for VGG-16 and VGG-19 were similar in both situations. However, when trained with batch sizes of 64 and 15 epochs, the models were slightly less accurate than the previous training (95.38% and 93.23%, respectively). Evidently, batch sizes of 32 and 15 epochs were suitable parameter values for the aforementioned models. However, ResNet-101 had a lower accuracy (42.46) compared with the other models when it was trained for 15 epochs with 32 batch sizes. Indeed, the accuracy increased when the batch size was increased to 64, yielding an accuracy rate of 76.24%.

Considering the noteworthy enhancement in accuracy observed in the model employing LSTM with ResNet-101 as the batch size increased from 32 to 64, the model was further examined by adjusting the epoch values to 50 and 100 to achieve compatibility with the specified batch sizes. The experimental results indicated that increasing the number of epochs to 50 and 100 improved the accuracy of the model. When trained with 100 epochs and 32 batch sizes, the model yielded the most accurate results at 92.45%. However, when trained with 100 epochs and 64 batch sizes, the accuracy decreased. The model demonstrated strong performance when applied to larger batch sizes and a greater number of epochs. The performance of the model using VGG-16 and VGG-19 was satisfactory when applied to small batch sizes and a reduced number of epochs. However, the LSTM with the ResNet-101 model exhibited a lower accuracy than the LSTM with the VGG-16 and VGG-19 models. The model employing LSTM in conjunction with VGG-16 required considerably shorter training times of 4.22 min and 27.08 min, respectively, as presented in Table 3.

The effectiveness of the proposed model was evaluated by combining the LSTM with the VGG-16 architecture for specific hyperparameters (15 epochs and a batch size of 32). The model was evaluated using a test dataset comprising 480 images obtained from the Celeb-DF dataset. The performance of the model was assessed using four key parameters: accuracy, precision, recall, and specificity, to determine its efficacy. The actual and predicted values for each quadrant of the confusion matrix are listed in Table 4.

The results showed that this model could effectively detect the fake faces with a high accuracy of 96.25%, recall of 93.04%, specificity of 99.20%, and precision of 99.07%, as presented in Table 5.

4.1. Discussion of Performance Compared to Relevant Research. This section focuses on seven studies that used similar techniques and examines their results, methods, and related research. In addition, the performance was assessed using the same public dataset, that is, Celeb-DF, as listed in Table 6.

The results showed that the proposed method had a higher accuracy than other relative models tested with the same standard dataset. Deep dual-level networks [37] had the nearest accuracy at the video level at 96.5%; however, it was slightly decreased to 94.8% at the frame level. Kaur et al. [39] proposed a C-LSTM model that combines a CNN and LSTM. The accuracy rate was 96.46% with a learning time of

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Table 4: Confusion matrix generated using VGG-16 + LSTM model with batch sizes of 32 and epochs of 15.

	Predicted fake	Predicted real
Actual fake	248 (TN)	2 (FP)
Actual real	16 (FN)	214 (TP)

Table 5: Evaluation results of the VGG-16 + LSTM model with batch sizes of 32 and epochs of 15.

Model	Accuracy (%)	Recall (%)	Specificity (%)	Precision (%)	Loss function (%)
VGG-16 + LSTM	96.25	93.04	99.20	99.07	6.54

Table 6: Performance results of the proposed methods and reviewed articles using the same Celeb-DF dataset.

Models	Accuracy (%)	Precision	Recall
C-LSTM [39]	96.46	99.62	93.91
Deep dual-level networks [37]	96.50	95.3	98.3
LSTM and Inception-ResNet [38]	91.48	N/A	N/A
CNN + VGG-19 [42]	94.65	N/A	N/A
CNN + ResNet-50 [42]	95.08	N/A	N/A
YOLO + CNN + XGBoost [41]	90.73	0.87	0.85
TD-3DCNN [43]	81.08	68.98	99.41
RCN [45]	76.25	N/A	N/A
R2Plus1D [45]	98.07	N/A	N/A
I3D [45]	92.28	N/A	N/A
MC3 [45]	97.49	N/A	N/A
R3D [45]	98.26	N/A	N/A
LSTM + VGG-16	**97.01**	**99.07%**	**93.04%**

The methods' performance values in this article are being bold to set them apart from methods' performance values in other articles.

56.91 min, which was 52.69 min more than that of the proposed method. In addition, it had a higher accuracy than the models reported by Rosli et al. [42] and Ismail et al. [41] that used a CNN in combination with other techniques such as XGBoost, VGG-16, and ResNet-50.

The predictive capabilities of various models were compared using recall and precision values. The proposed model had recall and precision values of 93.04 and 99.07, respectively. These values were compared with those in the study by Kaur et al. [39] that surpassed the values of the competing models. In conclusion, the proposed model had relatively high prediction accuracy for effectively differentiating between real and fake faces.

Yadav et al. [38] presented the LSTM and Inception-ResNet models that achieved 84.75% and 91.48% accuracy for 20 and 40 epochs, respectively. This is consistent with the test results presented in Table 3, where LSTM + ResNet-101 provides a higher accuracy when the epoch is configured at a high rate.

The proposed method outperformed other methods such as TD-3DCNN [43], RCN [45], and I3D [45] in terms of accuracy.

However, this model required a long learning time and its accuracy was slightly reduced when tested with actual detection. The model might be effective when it is extended to develop various applications, particularly those running on devices with lightweight processors such as mobile and embedded systems.

5. Conclusions

This study used LSTM with transfer learning models, including VGG-16, VGG-19, and ResNet-101, to determine a suitable deepfake detection model. Notably, the VGG-16 model had 97.01% accuracy and 8.60% loss function, with a total learning time of 4.22 min. This was higher than those of the other models that required the least time to learn. In addition, this study implemented LSTM with the VGG-16 model to detect fake faces using a standard dataset. The result showed that the model achieved 96.25% accuracy, 93.04% recall, 99.20% specificity, and 99.07% precision. In conclusion, this model can be practically implemented. In the future, this study will focus on reducing the learning time and increasing the detection accuracy by adding high-quality datasets to train the model. This model will continue to significantly improve cyber security and crime prevention apps to support a sustainable life.

Data Availability

The study compiled a collection of Celeb-DF video datasets available at https://paperswithcode.com/dataset/celeb-df. Previously reported deepfake VDO data were used to support this study and are available at https://paperswithcode.com/dataset/celeb-df.

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Disclosure

The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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